CORPORATE BANKING

FINFOLIO 2.0

Building the Future of Investing





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1. Limitation of Finfolio 1.0

Limited investment universe

- Primarily focused on a portfolio of Vietnamese stocks and the VN-Index benchmark

→ Difficult to assess diversification across asset classes

Lack of distribution in return and risk

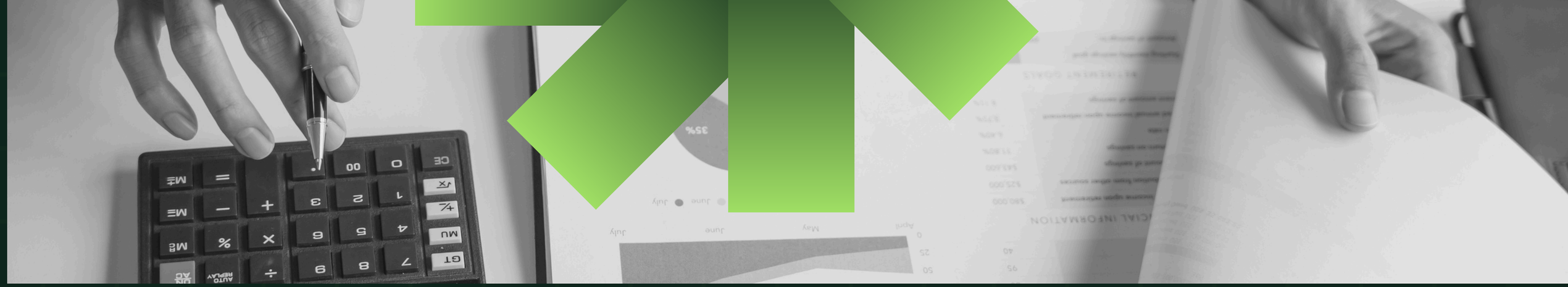
- Limited visibility into the sources of return, currency exposure and individual contributions to portfolio risk.

→ may mistakenly believe that an equal allocation of capital = an equal distribution of risk

Information heavy hard to navigate

- Too wide range of quantitative indicators, without explaination

→ User can be confused and find it hard to understand



2. Problem and target user

Target user

User Profile: Retail investors with foundational financial literacy.

Portfolio Scope: Self-managed multi-asset portfolios (Vietnam Equities + USD Commodity ETF proxies, e.g., GLD).

Four Portfolio-Level Deficits

Return Attribution Deficit: Cannot isolate returns from individual assets, sectors, or USD/VND FX.

Risk Concentration Blindness: Ignores covariance (e.g., 20% weight can drive 50% of total risk).

Benchmark Comparison Gap: Unaligned trading calendars when benchmarking against VN30.

Allocation Dilemma: Lacks quantitative proof for optimal risk-adjusted rebalancing.

SOLUTION ARCHITECTURE

3. User Data Flow

01 Input

- User holdings (asset, price, currency)
- Evaluation period
- Performance benchmark

02 Validate

Validate data symbol, data source, and the number of trading sessions

03 Attribute

Return and risk attribution (logic explained in previous pages)

04 Benchmark

Weight optimization process based on user-set constraints

05 Optimize

Weight optimization process based on user-set constraints (optional)

06 Simulate

Simulate 2000 scenarios based on historical data

07 **Decision**

Allows for decision record saving

3. User Flow & Defensive Architecture

Happy Path (Core Flow)

Portfolio Input -> Data Alignment -> Brief Report -> Reference Optimization -> Bootstrap Simulation -> Decision Log

Alternative Paths (Edge Handling)

Flexible Input: Accepts capital values (VND) or percentage weights.

Gap Fallback: Uses latest available session with warning banners.

Default Parameters: Pre-fills 3.0% annual risk-free rate.

Defensive Guardrails (2-Layer Security)

Invalid Tickers: Inline red alert; locks execution button.

Data Insufficiency: Blocks engine if < 60 sessions; hides optimizer if < 120 sessions.

Infeasible Constraints: Flags impossible weight limits (e.g., total cap < 100%).

DATA GOVERNANCE

4. How data are collected

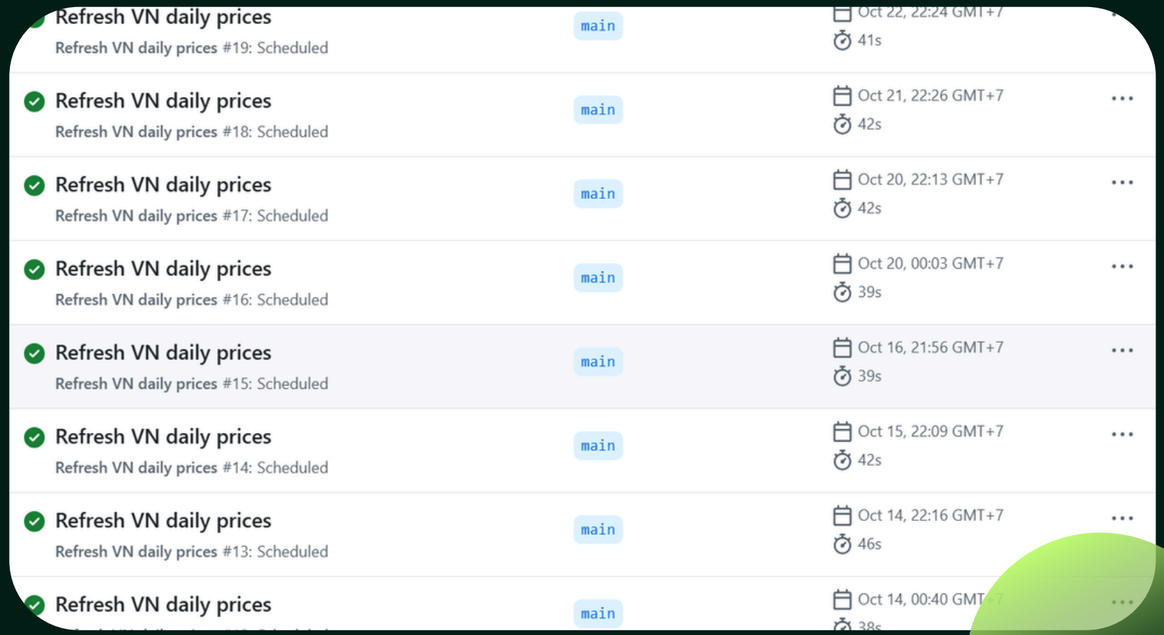
TradingView covers Vietnam:
Including 26 symbols (20 stocks, 4 ETFs, VN30, VN-Index

Yahoo covers USD-listed international assets
including 23 commodity ETFs & the USD/VND rate

Inner join
Keep only days on which all assets have a price.

Aligned price series

- Everything is converted to VND.
- Capital entered in USD is converted at the rate of the first common day T_0



Closed candles: Today's candle is ignored before 3:15 pm because it isn't a closing price yet.

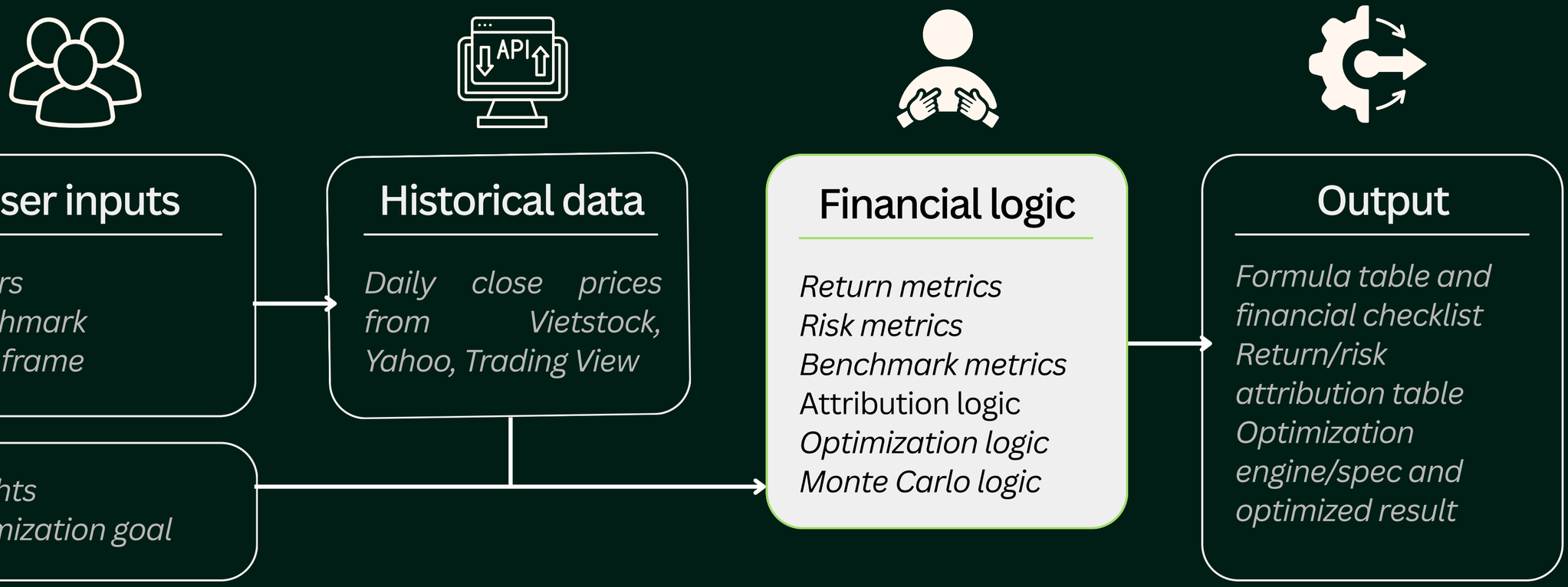
Snapshot: The 26 popular symbols are pre-downloaded every weekday after close. If a download fails, the old data keeps its old date.

Freshness: If the snapshot is missing the latest session, the system fetches it live from TradingView, so a late GitHub run doesn't affect the results.

Sector: Sectors come from TradingView. ETFs, commodities and crypto are marked "Not applicable".

5. Financial logic

General flow diagram



5. Financial logic

RETURN METRICS

Metric	Formula	Meaning	How to Analyze
Daily Return	$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$	Measures each asset's daily price change in VND terms.	Forms the basis for portfolio performance, risk and attribution analysis.
Portfolio Value	$V_t = \sum_i q_i P_{i,t}$	Tracks total portfolio value under fixed-quantity buy-and-hold.	Shows how the initial investment evolves without rebalancing.
Total Return	$R_p = \frac{V_T}{V_0} - 1$	Measures cumulative portfolio growth over the selected period.	Compare with the benchmark over the same historical period.
CAGR	$(\frac{Ending\ Value}{Beginning\ Value})^{1/Years} - 1$	Converts cumulative return into an equivalent annual growth rate.	Useful for comparing annualized growth across eligible periods.
Return Contribution	$C_i = w_{i,0} R_i$	Measures each asset's contribution to total portfolio return.	Identifies which assets or sectors drive gains and losses.
FX Attribution	$R_{VND} = R_L + R_{FX} + R_L R_{FX}$	Separates foreign-asset returns into local price, FX and interaction effects.	Reveals how currency movements affect VND-denominated performance.

5. Financial logic

RISK METRICS

Metric	Formula	Meaning	How to Analyze
Realized Volatility	$\sigma_{real} = s(r_p)\sqrt{252}$	Measures historical fluctuations in buy-and-hold portfolio returns.	Higher values indicate greater historical return variability.
Max Drawdown	$\min_t \left(\frac{V_t}{Peak_t} - 1\right)$	Measures the largest historical peak-to-trough portfolio decline.	Indicates the worst experienced decline within the selected period.
Marginal Contribution to Risk (MCR)	$MCR_i = \sqrt{252}\frac{(\Sigma w)_i}{\sqrt{w^\top \Sigma w}}$	Measures sensitivity of portfolio volatility to a marginal change in an asset's weight.	Identifies which positions increase or reduce portfolio risk at the margin.
Euler Risk Contribution (RC)	$RC_i = w_i MCR_i$	Allocates total portfolio volatility across individual assets.	Reveals whether an asset contributes disproportionately to portfolio risk relative to its weight.
VaR 95%	$VaR_{95} = Q_{0.95}(-r_p)$	Estimates the one-session loss threshold at 95% historical confidence.	Indicates a threshold exceeded in the worst historical tail.
CVaR / ES 95%	Avergae loss in the worst 5% of days	Measures average losses in the worst 5% of historical observations.	Reveals the severity of downside outcomes beyond the VaR threshold.

5. Financial logic

RISK ADJUSTED METRICS

Metric	Formula	Meaning	How to Analyze
Sharpe Ratio	$\frac{R_p - R_f}{\sigma_p}$	Measures excess return per unit of total volatility.	Higher values indicate better historical risk-adjusted performance.
Sortino Ratio	$\frac{R_p - R_f}{\sigma_d}$	Measures excess return relative to downside deviation.	Higher values suggest more compensation for harmful downside movements.
Calmar Ratio	$\frac{\text{Annualized } R_p}{\text{Maximum Drawdown}}$	Compares annualized growth with maximum historical drawdown.	A higher ratio indicates more historical growth per unit of drawdown.
Information Ratio	$\frac{R_p - R_b}{\text{Tracking Error}}$	Measures active return per unit of benchmark-relative risk.	Positive values indicate historical outperformance relative to active risk.

5. Financial logic

BENCHMARK COMPARING

Metric	Formula	Meaning	How to analyze
Beta	$\frac{\mathrm{Cov}(R_p, R_b)}{\mathrm{Var}(R_b)}$	Measures sensitivity to benchmark movement.	Beta > 1 means portfolio moves more strongly than benchmark. Beta < 1 means lower market sensitivity.
Alpha	$\alpha = R_p - [R_f + \beta \cdot (R_b - R_f)]$	Measures return beyond what is expected from benchmark risk exposure.	Alpha > 0 means the portfolio outperformed. Alpha < 0 means the portfolio underperformed after adjusting for beta risk.
Tracking Error	$\sqrt{\frac{\sum(R_{p,t} - R_{b,t} - \text{Active Return})^2}{n-1}}$	Measures how far portfolio returns deviate from benchmark returns.	Higher tracking error means the portfolio behaves less like the benchmark. It is not always bad; it depends whether active return is positive.
Information Ratio	$\frac{R_p - R_b}{\text{Tracking Error}}$	Measures excess return per unit of active risk.	Positive IR means active deviation was rewarded. Negative IR means the portfolio deviated from benchmark without enough return.

5. Financial logic

DIVERSIFICATION FUNCTIONS

Correlation từ -1 đến +1, trên return giá quy đổi VND cùng ngày. Thiếu biến động hiện "—"; tương quan lịch sử không cố định qua thời gian.

MÃ	FPT.VN	HPG.VN	GLD	BND	SLV	LQD	BTC-USD	VCB.VN	USO	VNM.VN
FPT.VN	1,00	0,43	-0,01	-0,06	0,03	-0,05	0,02	0,42	-0,02	0,27
HPG.VN	0,43	1,00	0,02	-0,01	0,08	-0,01	0,02	0,46	-0,05	0,41
GLD	-0,01	0,02	1,00	0,23	0,79	0,22	0,21	-0,00	0,04	0,03
BND	-0,06	-0,01	0,23	1,00	0,13	0,95	0,03	-0,08	-0,17	-0,01
SLV	0,03	0,08	0,79	0,13	1,00	0,15	0,23	0,04	0,01	0,08
LQD	-0,05	-0,01	0,22	0,95	0,15	1,00	0,09	-0,07	-0,15	-0,00
BTC-USD	0,02	0,02	0,21	0,03	0,23	0,09	1,00	0,04	-0,00	0,03
VCB.VN	0,42	0,46	-0,00	-0,08	0,04	-0,07	0,04	1,00	-0,00	0,40
USO	-0,02	-0,05	0,04	-0,17	0,01	-0,15	-0,00	-0,00	1,00	-0,04
VNM.VN	0,27	0,41	0,03	-0,01	0,08	-0,00	0,03	0,40	-0,04	1,00

Correlation matrix

5. Financial logic

ATTRIBUTION LOGIC

1. Instruments

Individual holding

Example: FPT.VN, VNM.VN,...

2. Asset class

Broad asset exposure

Example: Stock, ETF, Commodity, Cryto

3. Sector

Base on company sector as on Trading view

Example: Financial, Construction, Retail,...

4. Industry

Base on particular industry of instruments

Example: for stocks: In materials,: steel, plastic,...
for commodity: in metal: precious metal, industrial metal

5. Markets/commodity type

Base on the market that the instruments are trading or the type of commodity

Example: Vietnam, or international market

6. Currency

Denominations of assets

Example: USD, VND

5. Financial logic

OPTIMATION ASSUMPTIONS

Maximize Sharpe Ratio

Find the best historical return per unit of risk

Suitable for sers who want better risk-return efficiency

Minimize Volatility

Find the most stable historical portfolio

Suitable for users who prioritize lower risk and smoother returns

MARKOWITZ WITH FRANK–WOLFE OPTIMIZATION PROCESS

- Identify the direction for improvement
- Adjust allocation within constraints
- Repeat until the stopping criterion is met

→ More precise than Finfolio 1.0 with random search + hill climbing

KEY ASSUMPTIONS

01

Using VND-denominated daily returns

02

252 trading days/year
Risk-free rate: 3% p.a

03

Long-only, fully invested; asset & commodity exposure caps.

QUANTITATIVE ENGINE · OPTIMIZATION ENGINE

6. Frank-Wolfe Core Optimization

PROBLEM 1

Minimum Volatility

$$\min w^T \Sigma w$$

$\Sigma w_i = 1,\ 0 \le w_i \le Cap_i$

$\Sigma w_j \le Cap_comm\ (j \in Commodity)$

PROBLEM 2

Maximum Sharpe Ratio

$$\max (w^T \mu - r_f) / \sqrt{(w^T \Sigma w)} \cdot \sqrt{252}$$

r_f defaults to 3.0% per year

FRANK-WOLFE (CONDITIONAL GRADIENT)

20 ms

Solves the convex problem with box and unit-sum constraints in pure client-side JavaScript, removing the heuristic error of the old version.

CLAIM BOUNDARY

The result is a Reference Allocation based on historical data, not investment advice.

6. Frank-Wolfe Core Optimization

MONTE CARLO ASSUMPTIONS

- **Joint Moving-Block Bootstrap**

Resamples aligned 5-day return blocks, retaining observed cross-asset co-movements and some short-term dependence.

- **Simulation Settings**

2,000 paths × 252 sessions; 5-day blocks (default, changable)

Số phiên tương lai

252 phiên · khoảng 1 năm

Số đường mô phỏng

2.000

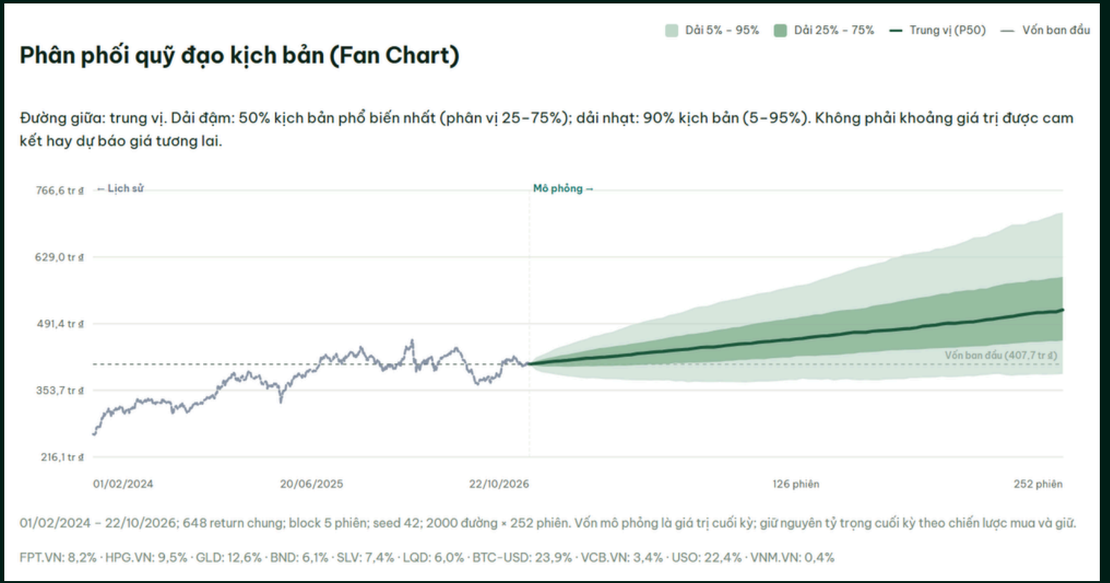
Seed · để tái lập kết quả

42

Lấy mẫu chung từng block 5 phiên cho mọi tài sản để giữ tương quan và một phần phụ thuộc ngắn hạn. Return đã quy đổi VND, gồm tác động FX. Không phải mô hình phân phối chuẩn hay dự báo giá.

Chạy mô phỏng

RESULTS



Các mốc phân vị cuối kỳ

MỐC PHÂN VỊ	GIÁ TRỊ VND	TĂNG / GIẢM	Ý NGHĨA XÁC SUẤT
Phân vị 5% (Kịch bản xấu)	387.279.949 đ	-5,0%	95% số đường mô phỏng đạt kết quả cao hơn mốc này. Mức kiểm thử sức chịu đựng.
Phân vị 25% (Thận trọng)	456.867.452 đ	+12,1%	75% số đường mô phỏng đạt kết quả cao hơn mốc này.
Phân vị 50% (Trung vị) CHUẨN	519.832.203 đ	+27,5%	Kịch bản cơ sở: một nửa số đường mô phỏng ở trên và một nửa ở dưới.
Phân vị 75% (Thuận lợi)	588.041.141 đ	+44,2%	25% kịch bản tốt nhất đạt từ mốc này trở lên.
Phân vị 95% (Rất thuận lợi)	720.724.307 đ	+76,8%	Chỉ 5% kịch bản hưng phấn nhất đạt tới mốc này.

PRODUCT SCOPE

7. Core MVP vs Final Release

	Core MVP	Final Release
Assets	Vietnamese stocks + 5 commodity ETFs, 2–3 positions	2–30 positions, adding international stocks/ETFs, 23 commodity ETFs, bond/real-estate ETFs, crypto
Optimization	Tried every weight combination in 1% steps	Frank-Wolfe algorithm (continuous optimization that respects limits) + Max-Sharpe
Risk	Volatility + Euler risk contribution	Adds Sortino, Calmar, VaR/ES 95%, Beta, Tracking error
Simulation	None	Bootstrap; 2,000 paths x 252 sessions





PRODUCT SCOPE

7. Core MVP vs Final Release

	Core MVP	Final Release
Data	Yahoo Finance only, temporary GLD benchmark	TradingView for Vietnam + Yahoo Finance for international, daily snapshot
Testing	Test scenarios T01–T06	85 automated tests
Decisions	Recorded in the session (lost when the tab closes)	Allows decision record retention (Google sign-in)

Out of scope Real-time prices, order execution, fees/taxes/dividends, exchange holiday calendar

VISUALIZATION DESIGN & USER EXPERIENCE (UI/UX)

8. Data Visualization & Design Tokens System

1 · COLOR & TYPE

Forest & Paper theme

#1F5A3F

#F5F6F1

Aa
Fraunces

Aa
Be Vietnam

Forest green on paper white builds trust and prevents eye strain.

30 self-hosted font variants with full Vietnamese support.

2 · NATIVE SVG CHART

PnL with dual-axis crosshair



Real VND values on the Y-axis, no abstract base-100 index.

The crosshair tracks X and Y; tooltips flip to avoid screen overflow.

3 · RESPONSIVE UX

Progressive disclosure

1 · Key metric cards

2 · Visual charts

3 · Detailed heatmaps

4 · Methodology

Creating a clear structure with a hierarchy, improving user flow.

TECHNICAL ARCHITECTURE & OPERATIONAL INFRASTRUCTURE

10. SYSTEM ARCHITECTURE

CLIENT BROWSER:

Main thread · UI

Vanilla JS ES Modules, smooth 60 fps interface

Web Worker

research-worker.mjs

Frank-Wolfe + 2,000 Monte Carlo runs

BACKEND

Serverless Python API

vn_data.py
global_data.py
classification.py

DATA PROVIDERS

TradingView
Yahoo

200 MS
initial page load, no React or Webpack

60 fps
UI stays smooth while heavy math runs in the Worker

2,000
Monte Carlo runs on a separate CPU thread

CI/CD
Vercel deploys and daily GitHub Actions data ingestion

11. QUALITY ASSURANCE & 85 AUTOMATED TESTS

Financial Identity (18 tests)

Euler risk decomposition and return alignment.

UI and Native SVG (12 tests)

Chart coordinates and currency formatting.

Optimization Solver (12 tests)

Frank-Wolfe convergence and weight limits.

Data Pipeline (35 tests)

Multi-market timezone alignment and fresh data endpoints.

Monte Carlo Simulation (8 tests)

Seed reproducibility and VaR bounds.

Core QA Principle: An automated regression suite blocks any code commit that violates financial logic, ensuring zero mathematical errors during live execution.

CONCLUSION

Finfolio 2.0 has successfully achieved its goals

- **A complete logical flow**
- **A solid financial math foundation**
- **An automated hybrid data pipeline**
- **A suite of 85 automated tests**

https://g09-finfolio.vercel.app/

Finfolio

